

Evaluation of Carbon Sink and Sequestration Potential of Trees within Hall - 8

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Summary

Trees act as important carbon sinks by absorbing atmospheric CO₂ and storing carbon in biomass. This study evaluates the **carbon stock and sequestration potential of trees within Hall-8, IIT Kanpur**, using field data on tree species, circumference, and height. These measurements were used to estimate above-ground and below-ground biomass, carbon stock, and corresponding CO₂ equivalent.

The study provides an assessment of the hall's existing tree-based carbon sink and its potential contribution to climate change mitigation. **The surveyed trees store approximately 1165.9143 kg of carbon, equivalent to 4275.01917 kg of CO₂, with an estimated annual sequestration potential of 464.21087 kg CO₂/year.**

1. Introduction

Carbon sequestration by vegetation is an important component of nature-based climate mitigation. Quantifying the carbon stored in trees provides a basis for understanding the contribution of green spaces to local carbon management.

In India, the **Forest Survey of India (FSI)** assesses forest carbon stocks using forest inventory data and biomass estimation methods. This study adopts a similar biomass-based approach at the institutional level to assess the carbon sink provided by trees within **Hall-8, IIT Kanpur**. **Source:** [Forest Survey of India – Carbon Reports](#)

Objectives

The main objectives of this study are:

1. To prepare an inventory of trees present within Hall - 8.
2. To estimate the above-ground biomass (AGB), below-ground biomass (BGB) of the surveyed trees.
3. To estimate the amount of carbon currently stored in the trees and convert the carbon stock into CO₂-equivalent terms.
4. To estimate the age of the surveyed trees and reconstruct their biomass and carbon stock of the subsequent year, using the growth equation derived.
5. To estimate the annual carbon sequestration rate by calculating the change in biomass and corresponding carbon stock over one year.
6. To identify the contribution of different tree species/tree-size classes to the carbon sink.

2. Methodology

The analysis was carried out in the following sequential steps:

The primary tree inventory data collected from Hall-8 were organized and checked for consistency. The recorded **circumference measurements were used to calculate diameter at breast height (DBH)**.

Circumference was measured at approximately **1.37 m above ground level**, following the standard convention used in forest inventory¹.

2.1 Primary Tree Inventory Data and Field Measurements

Data source: Primary field survey conducted within Hall - 8

Complete dataset: [Tree Inventory Dataset](#)

Block wise Distribution of Tree Biomass, Carbon Stock and CO₂ Equivalent

Specie Name	BW Block A-B	BW Block B-C	BW Block C-D	BW Block D-Mess	BW Block E-Mess	BW Block F - E	BW Block H - G	BW Hall office - Block H	Mess front	Outer road	Grand Total
Ailanthus excelsa	2	1		1						2	6
Artocarpus heterophyllus					1						1
Azadirachta indica		2	2	6	4	4	1		2	11	32
Bambusa sp.									2		2
Borassus flabellifer	1	2	2							1	6
Cassia fistula			1								1
Dalbergia sissoo	1	1	2	2						2	8
Delonix regia										1	1
Ficus religiosa										3	3
Mangifera indica	2	1	3	1	3	1	2	1		7	21
Manilkara zapota										1	1
Morus alba	3				1	1			1	1	7
Neolamarckia cadamba					1			1			2
Polyalthia longifolia										102	102
Psidium guajava	2	2	1	1		2	1		2		11
Syzygium cumini	1	1		2	3	1	1		1	4	14
Syzygium genus	1										1
Tectona grandis	1		3		1		2			1	8
Trapa natans									1		1
Grand Total	14	10	14	13	14	9	7	2	9	136	228

2.2 Estimation of Above-Ground Biomass

The above-ground biomass (AGB) accounts for the biomass of the stems and foliage to estimate the AGB of the surveyed trees was done using the generalized pantropical allometric model proposed by Chave et al. (2014). **Where D is the diameter at breast height (DBH), (ρ_i) stands for the density of the tree of specie (i) and H is the height of the tree.**

Relation between Circumferences C and D is $D = \frac{C}{\pi}$

$$Total\ AGB_{stems} = \sum_{n=1}^{228} (\rho_i \cdot D_i \cdot H_i)^{0.976}$$

Note: D, H is taken in the meters and Density (ρ_i) is taken in $\frac{kg}{m^3}$

For simplicity we used the direct result to calculate the AGB of the foliage as the 4% stems biomass²

$$Total\ AGB_{foliage} = 0.04 \cdot \sum_{n=1}^{228} (\rho_i \cdot D_i \cdot H_i)$$

The resulting **AGB represents dry biomass, expressed in kilograms (kg)**, and was converted into tones (t) where required for aggregate analysis.

2.3 Estimation of Below-Ground Biomass

Below-ground biomass (BGB), primarily representing the biomass of the root system, was not measured directly during the field survey. Instead, it was estimated from the calculated above-ground biomass using a below-ground-to-above-ground biomass ratio.

Based on empirical estimates, below-ground biomass (BGB) is assumed to be approximately 20% of above-ground biomass (AGB)³.

$$BGB = 0.2 \times Total\ AGB$$

The total biomass of each tree was then calculated as:

$$Total\ Biomass = Total\ AGB + BGB = Total\ AGB + 0.2 \times Total\ AGB$$

The estimated total biomass was subsequently used to calculate the carbon stock of the surveyed trees.

2.4 Estimation of Carbon Stock & Conversion of Carbon to CO₂ Equivalent

The carbon stock was estimated by multiplying biomass by the carbon fraction:

$$\text{Carbon Stock} = \text{Total Biomass} \times CF$$

A **carbon fraction of 0.47**, as established in **Indian forest-carbon methodology** and consistent with IPCC guidance, was applied to the estimated dry biomass to calculate the carbon stock⁴

To Convert the stored carbon as the corresponding amount of atmospheric CO₂

$$CO_2e = C \times \frac{44}{12}$$

The factor (44/12) arises from the molecular weights of CO₂ and carbon⁵.

$$\text{Thus, } 1 \text{ tonne } C \approx 3.667 \text{ tonnes } CO_2$$

2.5 Age-Based Estimation of Subsequent-Year's Biomass

To estimate the subsequent-year biomass and potential carbon sequestration, an age-based tree growth approach was adopted. Since direct measurements of annual tree growth were not available, a mathematical growth function was used to establish the relationship between tree age and circumference.

The **Chapman–Richards growth function**, a flexible forestry growth model, was fitted to the available age–circumference observations to characterize tree growth⁶:

$$C = K \cdot (1 - e^{-r \cdot A})^{\frac{1}{m}}$$

where C is circumference, A is age, and K , r , m are estimated growth parameters. The fitted relationship was used to estimate tree age and the annual circumference increment $t \left(\frac{dA}{dC} \right)_t$. The projected circumference was then converted to DBH and biomass to estimate subsequent-year carbon stock and potential CO₂ sequestration.

$$C_{t+1} = C_t + \left(\frac{dC}{dA} \right)_t ; D_{t+1} = \frac{C_{t+1}}{\pi}$$

$$\text{Chapman-Richards Model} \rightarrow C_{t+1} \rightarrow DBH_{t+1} \rightarrow Biomass_{t+1} \rightarrow CO_2 \text{ Sequestration}$$

2.6 Method of Estimation of Sequestration Potential

The current carbon stock represents the amount of carbon already stored in the existing biomass. To estimate **annual sequestration**, the change in biomass/carbon over time is required.

$$\text{Annual Carbon Sequestration} = \frac{C_{t+1} - C_t}{t + 1 - t} = C_{t+1} - C_t$$

Similarly, in CO₂-equivalent terms:

$$\text{Annual } CO_2 \text{ Sequestration} = \text{Annual Carbon Sequestration} \times \frac{44}{12}$$

Detailed calculations and supporting computations are provided in the Appendix.

3. Results and Discussion

The field survey identified **228 trees belonging to 19 species** within the study area. The estimated total above-ground biomass (AGB) was **1,805.431kg**, while below-ground biomass (BGB) was approximately **361.086 kg**, resulting in a total biomass of **2,166.517 kg**. Based on the estimated carbon fraction, the total carbon stock was approximately **1,018.263kgC**, equivalent to **3,733.631kgCO₂e** store in the tree biomass.

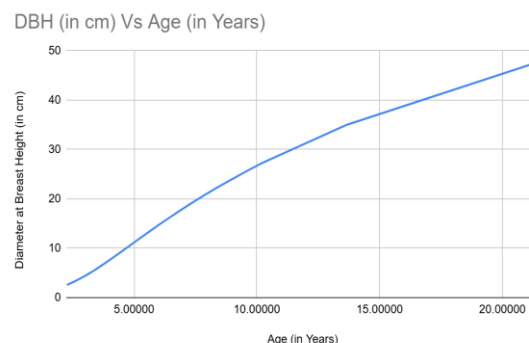
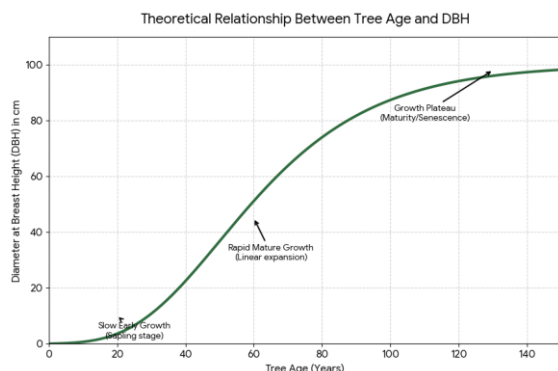
Furthermore, the estimated annual sequestration potential was **359.582kgC/year**, corresponding to approximately **1,318.467kgCO₂/year**. These findings indicate that the trees within **Hall-8 provide a measurable carbon sink and contribute to the removal and long-term storage of atmospheric CO₂ through biomass accumulation.**

3.1 Distribution of carbon stock among tree species

Species	COUNT Species	TB (in Kg)	TB (in Kg) at t + 1	Current carbon stock	Current carbon stock in CO ₂ e	Co ₂ sequestration potential
Ailanthus excelsa	6	36.169	49.430	16.999	62.331	22.853
Artocarpus heterophyllus	1	2.348	4.003	1.104	4.047	2.852
Azadirachta indica	32	306.847	420.577	144.218	528.799	195.995
Bambusa sp.	2	1.453	3.507	0.683	2.505	3.539
Borassus flabellifer	6	196.165	245.454	92.198	338.058	84.941
Cassia fistula	1	4.635	6.837	2.178	7.988	3.795
Dalbergia sissoo	8	127.782	163.861	60.058	220.211	62.176
Delonix regia	1	2.234	4.106	1.050	3.850	3.226
Ficus religiosa	3	389.780	475.663	183.197	671.722	148.004
Mangifera indica	21	99.260	143.712	46.652	171.059	76.604
Manilkara zapota	1	1.680	3.342	0.790	2.896	2.863
Morus alba	7	34.747	52.958	16.331	59.880	31.384
Neolamarckia cadamba	2	35.871	45.509	16.859	61.818	16.609
Polyalthia longifolia	102	647.475	924.025	304.313	1115.816	476.588
Psidium guajava	11	24.566	44.392	11.546	42.336	34.166
Syzygium cumini	14	168.921	225.244	79.393	291.107	97.064
Syzygium genus	1	3.628	5.565	1.705	6.252	3.338
Tectona grandis	8	74.387	101.116	34.962	128.193	46.063
Trapa natans	1	8.568	12.285	4.027	14.765	6.407
Grand Total	228	2166.517	2931.586	1018.263	3733.631	1318.468

3.2 Relationship between DBH Growth with time

The theoretical growth relationship plot generated from the Chapman–Richards growth function was compared with the observed age–DBH data from Hall-8 to assess the consistency of the estimated tree growth pattern⁷.



Limitations: The estimated carbon stock depends on the accuracy of DBH and height measurements, the selected allometric equation, wood-density values, below-ground biomass assumptions, and the applied carbon fraction. In addition, trees with a circumference below 8 cm were excluded from the analysis. Therefore, the estimated carbon stock should be interpreted as an approximation rather than an exact measure of carbon stored in the trees.

4. Conclusion

This study assessed the carbon storage and estimated sequestration potential of 228 trees belonging to 19 species in Hall-8. The surveyed trees contained approximately 2,166.62 kg of total biomass, corresponding to 1,018.26 kg of carbon stock and 3,733.83 kg CO₂-equivalent stored in their biomass. The estimated annual sequestration potential was approximately 359.58 kg C yr⁻¹, equivalent to 1,318.49 kg CO₂ yr⁻¹.

Carbon storage was not evenly distributed among species. *Polyalthia longifolia*, which was also the most abundant species (102 trees), contributed the largest carbon stock (304.31 kg C) and estimated sequestration potential (476.59 kg CO₂ yr⁻¹). Other important contributors included *Ficus religiosa*, *Azadirachta indica*, and *Syzygium cumini*, reflecting the influence of both tree abundance and individual tree biomass on the overall carbon stock.

The study therefore establishes a baseline estimate of carbon stock and potential sequestration for Hall-8. However, the sequestration values represent model-based estimates rather than directly observed annual changes in biomass. Re-measuring the same trees after one or more years would allow the estimated growth to be validated and provide a more robust measure of the actual annual carbon sequestration rate.

I would not call 1,318.49 kg CO₂/year the "actual sequestration" in the conclusion. Call it: **Estimated annual CO₂ sequestration potential**.

5. Statement on Use of AI Tools

AI Tool Usage Statement:

Artificial Intelligence (AI) tools were used in the preparation of this report for **literature-search assistance / understanding carbon-sequestration concepts / organizing the report structure / checking mathematical formulations / improving language and presentation**. The field data, measurements, assumptions, calculations, and interpretation of the results were **[collected/performed/verified] by me**. AI-generated information was cross-checked with relevant academic and institutional sources before being incorporated into the report.

6. References

1. Puc-Kauil, R. et al. (2020). *iForest*, 13, 165–174. <https://doi.org/10.3832/ifor3167-013>
2. AGB was estimated using the generalized pantropical allometric equation proposed by Chave et al. (2014).
3. MacDicken, K. G. (1997). *A Guide to Monitoring Carbon Storage in Forestry and Agroforestry Projects*. Winrock International.
4. IPCC (2006). *2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 4: Agriculture, Forestry and Other Land Use*. Institute for Global Environmental Strategies (IGES), Japan.
5. IPCC (2006). *2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2: Energy*. Institute for Global Environmental Strategies (IGES), Japan.
6. Richards, F. J. (1959). A flexible growth function for empirical use. *Journal of Experimental Botany*, 10(2), 290–300. <https://doi.org/10.1093/jxb/10.2.290>
7. **Richards, F. J. (1959). A flexible growth function for empirical use. *Journal of Experimental Botany*, 10(2), 290–300.**

Appendix A: Detailed Calculations for Tree Growth and Carbon Sequestration

1. Chapman–Richards's growth function

$$C = K(1 - e^{-rA})^{\frac{1}{m}}$$

where C represents tree circumference (cm), A represents tree age (years), and K, r, and m are the estimated growth parameters.

2. Circumference growth rate

Differentiating the growth function with respect to age gives the annual circumference growth rate:

$$\frac{dC}{dA} = \frac{Kr}{m} e^{-rA} (1 - e^{-rA})^{\frac{1}{m}-1}$$

3. Subsequent-year circumference

The circumference in the subsequent year was estimated by adding the annual growth increment to the current circumference:

$$C_{t+1} = C_t + \left(\frac{dC}{dA} \right)_t = C_t + \frac{Kr}{m} e^{-rA} (1 - e^{-rA})^{\frac{1}{m}-1}$$

4. Conversion of circumference to DBH

The projected circumference was converted to diameter at breast height (DBH) using:

$$DBH_{t+1} = \frac{C_{t+1}}{\pi}$$

5. Subsequent-year biomass

The projected DBH was then used in the selected allometric equation to estimate subsequent-year above-ground biomass:

$$AGB_{t+1} = 0.0673(\rho DBH_{t+1}^2 H)^{0.976}$$

6. Biomass increment

The annual biomass increment was calculated as

$$\Delta Biomass = Biomass_{\{t+1\}} - Biomass_t$$

7. Carbon sequestration

The corresponding increase in carbon stock was estimated using the carbon fraction:

$$\Delta C = \Delta Biomass \times CF$$

8. CO₂ sequestration potential

Finally, the annual carbon sequestration was converted into its CO₂ equivalent using the molecular-weight ratio of carbon dioxide to carbon:

$$\Delta CO_2 = \Delta C \times \frac{44}{12}$$

This sequence provides the basis for estimating the **potential annual CO₂ sequestration** from the projected increase in tree biomass.